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Missing Value Imputation for Mixed Data via Gaussian Copula

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ABSTRACT

Missing data imputation forms the first critical step of many data analysis pipelines. The challenge is greatest for mixed data sets, including real, Boolean, and ordinal data, where standard techniques for imputation fail basic sanity checks: for example, the imputed values may not follow the same distributions as the data. This paper proposes a new semiparametric algorithm to impute missing values, with no tuning parameters. The algorithm models mixed data as a Gaussian copula. This model can fit arbitrary marginals for continuous variables and can handle ordinal variables with many levels, including Boolean variables as a special case. We develop an efficient approximate EM algorithm to estimate copula parameters from incomplete mixed data. The resulting model reveals the statistical associations among variables. Experimental results on several synthetic and real datasets show the superiority of our proposed algorithm to state-of-the-art imputation algorithms for mixed data.

CCS CONCEPTS

• **Mathematics of computing** → **Expectation maximization; Maximum likelihood estimation**; • **Computing methodologies** → **Learning latent representations**.

KEYWORDS

mixed data, ordinal data, Gaussian copula, missing values, imputation

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1 INTRODUCTION

Mixed data sets — those that include real, Boolean, and ordinal data — are a fixture of modern data analysis. Ordinal data is particularly common in survey datasets. For example, Netflix users rate movies on a scale of 1–5. Social surveys may roughly bin respondents' income or level of education as an ordinal variable, and ordinal Likert scales measure how strongly a respondent agrees with certain stated opinions. Binary variables may be considered a special case of an ordinal with two levels. Health data often contains ordinals

that result from patient surveys or from coarse binning of continuous data into, e.g., cancer stages 0–IV or overweight vs obese patients. In all of these settings, missing data is endemic due to non-response and usually represents a large proportion of the dataset. Missing value imputation generally precedes other analysis, since most machine learning algorithms require complete observations. Imputation quality can strongly influence subsequent analysis.

To exploit the information in mixed data, imputation must account for the interaction between continuous and ordinal variables. Unfortunately, the joint distribution of mixed data can be complex. Existing parametric models are either too restrictive [18] or require priori knowledge of the data distribution [35]. Nonparametric methods, such as MissForest [29], based on random forests, and imputeFAMD [2], based on principal components analysis, tend to perform better. However, these two methods treat ordinal data as categorical, losing valuable information about the order. Further, they can only afford a limited number of categories.

It is tempting, but dangerous, to treat ordinal data with many levels as continuous. For example, the ordinal variable “Weeks Worked Last Year” from the General Social Survey dataset takes 48 levels, but 74% of the population worked either 0 or 52 weeks. Imputation that treats this variable as continuous (e.g., imputing with the mean) works terribly! As another example, consider using low rank matrix completion [5, 16, 20, 25] to impute missing entries in a movie rating datasets using a quadratic loss. This loss implicitly treats ratings encoded as 1–5 as numerical values, so the difference between ratings 3 and 4 is the same as that between ratings 4 and 5. Is this true? How could we tell?

A more sensible (and powerful) model treats ordinal data as generated by thresholding continuous data, as in [26, 27]. Figure 1 illustrates how correlations can be garbled by treating such data as continuous. Our work builds on this intuition to model mixed data through the Gaussian copula model [8, 9, 14, 19], which assumes the observed vector is generated by transforming each marginal of a latent normal vector. Under this model, we associate each variable (both ordinal and continuous) with a latent normal variable. Each ordinal level corresponds to an interval of values of the corresponding latent normal variable.

We propose an efficient EM algorithm to estimate a Gaussian copula model with incomplete mixed data and show how to use this model to impute missing values. Our method outperforms many state-of-the-art imputation algorithms for various real datasets including social survey data (whose columns have a varying number of ordinal levels), movie rating data (high missing ratio), music tagging data (binary data), etc. The proposed method has several advantages: the method has no hyper-parameters to tune and is invariant to coordinate-wise monotonic transformations in the data. Moreover, the fitted copula model is interpretable and can reveal statistical associations among variables, which is useful for social science applications. To our best knowledge, our proposed

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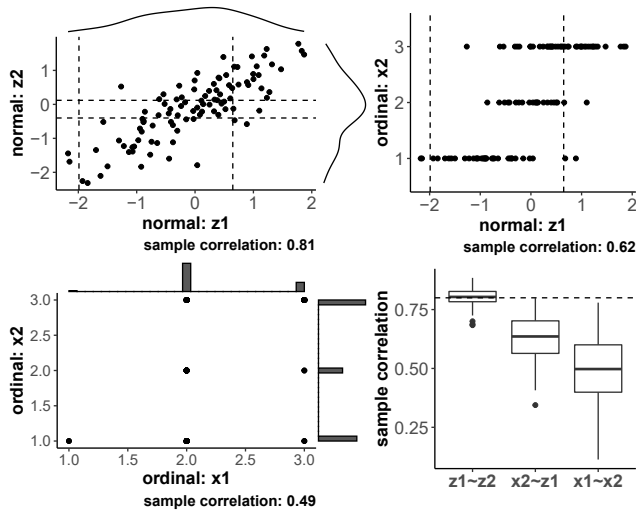


Figure 1: Draw (z_1, z_2) from a binormal with correlation 0.8. Discretize z_1 to x_1 , z_2 to x_2 on random cutoffs. Top two and bottom left panels plot one repetition. Dashed lines mark the cutoffs. Bottom right panel plots the sample correlation over 100 repetitions. Dashed line marks the truth.

algorithm is the first frequentist approach to fit the Gaussian copula model with incomplete mixed data. Moreover, it is much faster than the existing Bayesian MCMC algorithm for the same model [14]; given the same time budget, our method produces substantially more accurate estimates.

1.1 RELATED WORK

Gaussian Copula for Mixed Data. Modeling mixed data with the Gaussian copula model has been studied using both frequentist approaches [8, 9] and Bayesian approaches [6, 14, 22]. In [6, 22], the authors further assume the latent normal vector is generated from a factor model. When all variables are ordinal, the Gaussian copula model is equivalent to the probit graphical model [11]. However, all these previous work focuses on model estimation and theoretical properties of the estimators, and has overlooked the potential of these models for missing value imputation.

In fact, the frequentist parameter estimation methods proposed [8, 9, 11] assume complete data; so these methods cannot perform imputation. Among Bayesian approaches, MCMC algorithms [6, 14, 22] can fit the copula model with incomplete data and impute missing values. However, to use these models, one must select the number of factors for the models in [6, 22]. The sensitivity of these models to this parameter makes it a poor choice in practice for missing value imputation.

The implementation of [14] is still the best method available to fit a Gaussian copula model for incomplete mixed data. Hollenbach et al. [15] provides an important case study of this method for use in multiple imputation with an application to sociological data analysis. However, the method is slow and sensitive: the burn-in and sampling period must be carefully chosen for MCMC to converge, and many iterations are often required, so the method

does not scale to even moderate size data, which limits its use in practice. Our model matches that of [14], but our EM algorithm runs substantially faster.

Low rank matrix completion. The generalized low rank models framework [32] handles missing values imputation for mixed data using a low rank model with appropriately chosen loss functions to ensure proper treatment of each data type. However, choosing the right loss functions for mixed data is challenging. A few papers share our motivation: for example, early papers by Rennie and Srebro [26, 27] proposed a thresholding model to generate ordinals from real low rank matrices. Ganti et al. [10] estimate monotonic transformations of a latent low rank matrix, but the method performs poorly in practice. Anderson-Bergman et al. [1] posits that the mixed data are generated by marginally transforming the columns of the sum of a low rank matrix and isotropic Gaussian noise. While their marginal transformation coincides with the Gaussian copula model, their setup greatly differs in that it cannot identify the correlations between variables.

While low rank matrix completion methods scale well to large datasets, the low rank assumption is too weak to generalize well on long skinny datasets. Hence low rank methods tend to work well on “square-ish” datasets ($n \sim p$) [33], while the copula methods proposed here work better on long, skinny datasets.

2 NOTATION

Define $[p] = \{1, \dots, p\}$ for $p \in \mathbb{Z}$. Let $\mathbf{x} = (x_1, \dots, x_p) \in \mathbb{R}^p$ be a random vector. We use $\mathbf{x}_I$ to denote the subvector of $\mathbf{x}$ with entries in subset $I \subset [p]$. Let $\mathcal{M}, \mathcal{C}, \mathcal{D} \subset [p]$ denote missing, observed continuous, and observed discrete (or ordinal) dimensions, respectively. The observed dimensions are $\mathcal{O} = \mathcal{C} \cup \mathcal{D}$, so $\mathbf{x} = (\mathbf{x}_{\mathcal{C}}, \mathbf{x}_{\mathcal{D}}, \mathbf{x}_{\mathcal{M}}) = (\mathbf{x}_{\mathcal{O}}, \mathbf{x}_{\mathcal{M}})$.

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a matrix whose rows correspond to observations and columns to variables. We refer to the i -th row, j -th column, and (i, j) -th element as $\mathbf{x}^i$, $\mathbf{X}_j$ and x_j^i , respectively.

We say random variables $x = y$ and random vectors $\mathbf{x} = \mathbf{y}$ if their cumulative distribution functions (CDF) match. The elliptope $\mathcal{E} = \{Z \geq 0 : \text{diag}(Z) = 1\}$ is the set of correlation matrices.

3 GAUSSIAN COPULA

The Gaussian copula models complex multivariate distributions through transformations of a latent Gaussian vector. We call a random variable $x \in \mathbb{R}$ continuous when it is supported on an interval. We can match the marginals of any continuous random vector $\mathbf{x}$ by applying a strictly monotone function to a random vector $\mathbf{z}$ with standard normal marginals. Further, the required function is unique, as stated in Lemma 1.

Lemma 1. Suppose $\mathbf{x} \in \mathbb{R}^p$ is a continuous random vector with CDF F_j for each coordinate $j \in [p]$, and $\mathbf{z} \in \mathbb{R}^p$ is a random vector with standard normal marginals. Then there exists a unique elementwise strictly monotone function $\mathbf{f}(\mathbf{z}) := (f_1(z_1), \dots, f_p(z_p))$ such that

$$x_j = f_j(z_j) \quad \text{and} \quad f_j = F_j^{-1} \circ \Phi, \quad j \in [p], \quad (1)$$

where Φ is the standard normal CDF.

All proofs appear in the supplementary materials. Notice the functions $\{f_j\}_{j=1}^p$ in Eq. (1) are strictly monotone, so their inverses exist. Define $\mathbf{f}^{-1} = (f_1^{-1}, \dots, f_p^{-1})$. Then $\mathbf{z} = \mathbf{f}^{-1}(\mathbf{x})$ has standard normal marginals, but the joint distribution of $\mathbf{z}$ is not uniquely determined. The Gaussian copula model (or equivalently nonparametric distribution [19]) further assumes $\mathbf{z}$ is jointly normal.

Definition 1. We say a continuous random vector $\mathbf{x} \in \mathbb{R}^p$ follows the Gaussian copula $\mathbf{x} \sim \text{GC}(\Sigma, \mathbf{f})$ with parameters Σ and $\mathbf{f}$ if there exists a correlation matrix Σ and elementwise strictly monotone function $\mathbf{f} : \mathbb{R}^p \rightarrow \mathbb{R}^p$ such that $\mathbf{f}(\mathbf{z}) = \mathbf{x}$ for $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \Sigma)$.

This model is semiparametric: it comprises nonparametric functions $\mathbf{f}$ and parametric copula correlation matrix Σ . The monotone $\mathbf{f}$ establishes the mapping between observed $\mathbf{x}$ and latent normal $\mathbf{z}$, while Σ fully specifies the distribution of $\mathbf{z}$. Further, the correlation Σ is invariant to elementwise strictly monotone transformation of $\mathbf{x}$. Concretely, if $\mathbf{x} \sim \text{GC}(\Sigma, \mathbf{f})$ and $\mathbf{y} = \mathbf{g}(\mathbf{x})$ where $\mathbf{g}$ is elementwise strictly monotone, then $\mathbf{y} \sim \text{GC}(\Sigma, \mathbf{f} \circ \mathbf{g}^{-1})$. Thus the Gaussian copula separates the multivariate interaction Σ from the marginal distribution $\mathbf{f}$.

When f_j is strictly monotone, x_j must be continuous. On the other hand, when f_j is monotone but not strictly monotone, x_j takes discrete values in the range of f_j and can model ordinals. Thus for ordinals, f_j will not be invertible. For convenience, we define a set-valued inverse $f_j^{-1}(x_j) := \{z_j : f_j(z_j) = x_j\}$. When the ordinal x_j has range $[k]$, Lemma 2 states that the only monotone function f_j mapping continuous z_j to x_j is a cutoff function, defined for some parameter $S \subset \mathbb{R}$ as

$$\text{cutoff}(z; S) := 1 + \sum_{s \in S} \mathbb{1}(z > s) \text{ for } z \in \mathbb{R}.$$

Lemma 2. Suppose $x \in \mathbb{R}$ is an ordinal random variable with range $[k]$ and probability mass function $\{p_l\}_{l=1}^k$ and $z \in \mathbb{R}$ is a continuous random variable with CDF F_z . Then $f = \text{cutoff}(z; S)$ is the unique monotone function f that satisfies $x = f(z)$, where $S = \{s_l = F_z^{-1}(\sum_{t=1}^l p_t) : l \in [k-1]\}$.

For example, in recommendation system we can think of the discrete ratings as obtained by rounding some ideal real valued score matrix. The rounding procedure amounts to apply a cutoff function. See Figure 2 for an example of cutoff function.

To extend the Gaussian copula to mixed data, we simply specify that f_j is strictly monotone for $j \in \mathcal{C}$ and that f_j is a cutoff function for $j \in \mathcal{D}$. As before, the correlation Σ remains invariant to elementwise strictly monotone transformations. The main difference is that while $f_j^{-1}(x_j)$ is a single number when $j \in \mathcal{C}$ is continuous, it is an interval when $j \in \mathcal{D}$ is discrete. See Figure 3 for illustration.

4 IMPUTATION

So far we have introduced a very flexible model for mixed data. Our interest is to investigate missing value imputation under this model. Concretely, suppose the data matrix $\mathbf{X}$ has rows $\mathbf{x}^1, \dots, \mathbf{x}^n$ i.i.d. $\text{GC}(\Sigma, \mathbf{f})$ and $\mathbf{x}^i = (\mathbf{x}_{\mathcal{C}_i}^i, \mathbf{x}_{\mathcal{D}_i}^i, \mathbf{x}_{\mathcal{M}_i}^i) = (\mathbf{x}_{\mathcal{O}_i}^i, \mathbf{x}_{\mathcal{M}_i}^i)$ for $i \in [n]$. we first estimate $\hat{\Sigma}$ and $\hat{\mathbf{f}}$ using observation $\{\mathbf{x}_{\mathcal{O}_i}^i\}_{i=1}^n$ and then impute missing values $\{\mathbf{x}_{\mathcal{M}_i}^i\}_{i=1}^n$ using $\hat{\Sigma}$, $\hat{\mathbf{f}}$ and observation $\{\mathbf{x}_{\mathcal{O}_i}^i\}_{i=1}^n$.

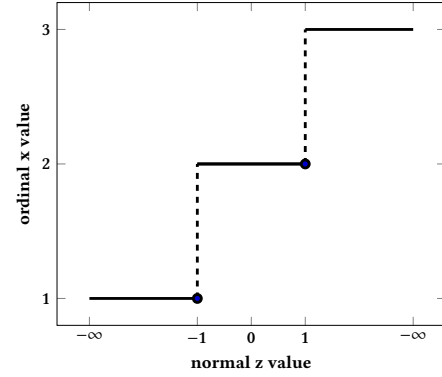


Figure 2: Cutoff function $f(\cdot)$ with cutoffs $\{-1, 1\}$ maps continuous z to ordinal $x \in \{1, 2, 3\}$.

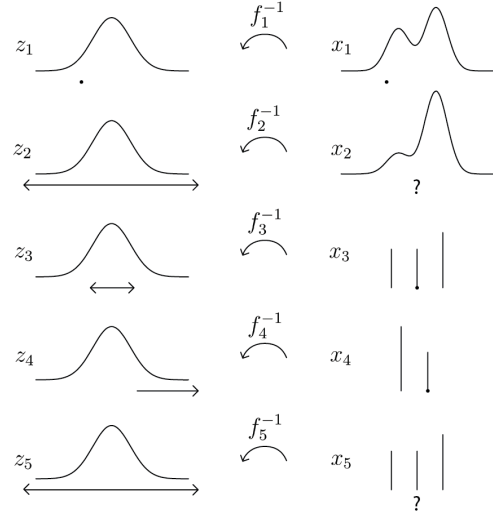


Figure 3: An example of 5 dimensional Gaussian copula model. For observed continuous x_1 , the corresponding z_1 takes a fixed value. For observed ordinal x_3 and x_4 , the corresponding z_3 and z_4 take values from an interval. For missing continuous x_2 and missing ordinal x_5 , the corresponding z_2 and z_5 can take any value.

In this section we first show how to impute the missing values with given estimates $\hat{\mathbf{f}}$ and $\hat{\Sigma}$. The estimation for $\mathbf{f}$ appears in Section 5, and the estimation for Σ appears in Section 6. The missing completely at random (MCAR) assumption is needed to consistently estimate $\mathbf{f}$. If the true $\mathbf{f}$ is known, the missing at random (MAR) assumption suffices to consistently estimate Σ . We discuss this issue further later in the paper.

For the latent normal vector $\mathbf{z}^i$ satisfying $\mathbf{x}^i = \mathbf{f}(\mathbf{z}^i)$, $\mathbf{z}^i$ follows truncated normal distribution. Define $\mathbf{f}_I = (f_j)_{j \in I}$ for $I \subset [p]$ and $f_j^{-1}(x_j^i) = \mathbb{R}$ for $j \in \mathcal{M}_i$. In observed continuous dimensions $\mathcal{C}_i$, $\mathbf{z}_{\mathcal{C}_i}^i$ reduces to the point $\mathbf{f}_{\mathcal{C}_i}^{-1}(\mathbf{x}_{\mathcal{C}_i}^i)$. In observed ordinal dimensions $\mathcal{D}_i$, $\mathbf{z}_{\mathcal{D}_i}^i$ lies in the Cartesian product of intervals $\mathbf{f}_{\mathcal{D}_i}^{-1}(\mathbf{x}_{\mathcal{D}_i}^i)$. There is

no constraint in missing dimension $\mathcal{M}_i$. It is natural to impute $\mathbf{x}_{\mathcal{M}_i}^i$ by mapping the conditional mean of $\mathbf{z}_{\mathcal{M}_i}^i$ through the marginals $\mathbf{f}_{\mathcal{M}_i}$, summarized in Algorithm 1.

Algorithm 1 Imputation via Gaussian Copula

Input: observation $\{\mathbf{x}_{O_i}^i\}_{i=1}^n$, parameters estimate $\hat{\mathbf{f}}^{-1}$ and $\hat{\Sigma}$.

- (1) Compute constraints $\mathbf{z}_{O_i}^i \in \hat{\mathbf{f}}_{O_i}^{-1}(\mathbf{x}_{O_i}^i), i \in [n]$.
- (2) For $i = 1, \dots, n$,
 - Impute $\hat{\mathbf{z}}_{\mathcal{M}_i}^i = \mathbb{E}[\mathbf{z}_{\mathcal{M}_i}^i | \mathbf{z}_{O_i}^i \in \hat{\mathbf{f}}_{O_i}^{-1}(\mathbf{x}_{O_i}^i), \hat{\Sigma}]$.
 - Impute $\hat{\mathbf{x}}_{\mathcal{M}_i}^i = \hat{\mathbf{f}}_{\mathcal{M}_i}(\hat{\mathbf{z}}_{\mathcal{M}_i}^i)$.

Output: $\hat{\mathbf{x}}_{\mathcal{M}_i}^i$ for $i \in [n]$.

While most applications require just a single imputation, multiple imputations are useful to describe the uncertainty due to imputation. Our method also supports multiple imputation: in step (2) of Algorithm 1, replace the conditional mean imputation with conditional sampling and then impute $\hat{\mathbf{x}}_{\mathcal{M}_i}^i$ for each sample. The conditional sampling consists of two steps: (1) sample the truncated normal $\mathbf{z}_{O_i}^i$ conditional on $\mathbf{x}_{O_i}^i$ and $\hat{\Sigma}$; (2) sample the normal $\mathbf{z}_{\mathcal{M}_i}^i$ conditional on $\mathbf{z}_{O_i}^i$ and $\hat{\Sigma}$. Efficient sampling methods have been proposed [23] for multivariate truncated normal distribution.

5 MONOTONIC FUNCTION ESTIMATION

To map between $\mathbf{x}$ and $\mathbf{z}$, we require both $\mathbf{f}^{-1}$ and $\mathbf{f}$. It is easier to directly estimate $\mathbf{f}^{-1}$. For $j \in C$, we have $\hat{f}_j^{-1} = \Phi^{-1} \circ F_j$, as shown in Eq. (1). While the true CDF F_j is usually unavailable, it is natural to estimate it by the empirical CDF of $\mathbf{X}_j$ on the observed entries, denoted as $\hat{F}_j$. We use the following estimator:

$$\hat{f}_j^{-1}(\mathbf{x}_j^i) = \Phi^{-1} \left(\frac{n}{n+1} \hat{F}_j(\mathbf{x}_j^i) \right). \quad (2)$$

The scale constant $n/(n+1)$ ensures the output is finite. MCAR assumption guarantees the observed entries of $\mathbf{X}_j$ are from the distribution of F_j . Consider a case when MCAR is violated: an entry is observed if and only if it is smaller than a constant c , then the observed entries are actually from the distribution $\tilde{F}_j$:

$$\tilde{F}_j(\mathbf{x}_j) = \begin{cases} F_j(\mathbf{x}_j)/F_j(c), & \text{when } \mathbf{x}_j \leq c \\ 1, & \text{when } \mathbf{x}_j > c \end{cases}.$$

Thus we assume MCAR in this section. This assumption may be relaxed to MAR or even missing not at random by carefully modeling F_j or the missing mechanism. We leave that to our future work. Lemma 3 shows this estimator converges to $\hat{f}_j^{-1}$ in sup norm on the observed domain.

Lemma 3. Suppose the continuous random variable $x \in \mathbb{R}$ with CDF F_x and normal random variable $z \in \mathbb{R}$ satisfy $f(z)=x$ for a strictly monotone f . Given $x^1, \dots, x^n \stackrel{i.i.d.}{\sim} F_x$, $m = \min_i x^i$, and $M = \max_i x^i$, the inverse $\hat{f}^{-1}$ defined in Eq. (2) satisfies

$$\mathbb{P} \left(\sup_{m \leq x \leq M} |\hat{f}^{-1}(x) - f^{-1}(x)| > \epsilon \right) \leq 2e^{-c_1 n \epsilon^2},$$

for any ϵ in $a_1 n^{-1} < \epsilon < b_1$, where $a_1, b_1, c_1 > 0$ are constants depending on $F_x(m)$ and $F_x(M)$.

For an ordinal variable $j \in \mathcal{D}$ with k levels, $f_j(z_j) = \text{cutoff}(z_j; \mathbf{S}^j)$. Since $\mathbf{S}^j$ is determined by the probability mass function $\{p_l^j\}$ of x_j , we may estimate cutoffs $\hat{\mathbf{S}}^j$ as a special case of Eq. (2) by replacing p_l^j with its sample mean:

$$\mathbf{S}^j = \left\{ \Phi^{-1} \left(\frac{\sum_{i=1}^{n_j} \mathbb{1}(x_j^i \leq l)}{n_j + 1} \right), l \in [k-1] \right\}. \quad (3)$$

Lemma 4 shows that $\hat{\mathbf{S}}^j$ consistently estimates $\mathbf{S}^j$.

Lemma 4. Suppose the ordinal random variable $x \in [k]$ with probability mass function $\{p_l\}_{l=1}^k$ and normal random variable $z \in \mathbb{R}$ satisfy $f(z) = \text{cutoff}(z; \mathbf{S})=x$. Given samples $x^1, \dots, x^n \stackrel{i.i.d.}{\sim} \{p_l\}_{l=1}^k$, the cutoff estimate $\hat{\mathbf{S}}$ from Eq. (3) satisfies

$$\mathbb{P} \left(\|\hat{\mathbf{S}} - \mathbf{S}\|_1 > \epsilon \right) \leq 2^k e^{-c_2 n \epsilon^2 / (k-1)^2},$$

for any ϵ in $(k-1)a_2 n^{-1} < \epsilon < (k-1)b_2$, where $a_2, b_2, c_2 > 0$ are constants depending on $\{p_1, p_k\}$.

6 COPULA CORRELATION ESTIMATION

We first consider maximum likelihood estimation (MLE) for Σ with complete continuous observation, then generalize the estimation method to incomplete mixed observation.

6.1 Complete Continuous Observations

We begin by considering continuous, fully observed data: $\mathcal{D} = \mathcal{M} = \emptyset$. The density of the observed variable $\mathbf{x}$ is

$$p(\mathbf{x}; \Sigma, \mathbf{f}) d\mathbf{x} = \phi(\mathbf{z}; \Sigma) d\mathbf{z},$$

where $\mathbf{z} = \mathbf{f}^{-1}(\mathbf{x})$, $d\mathbf{z} = \left| \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right| d\mathbf{x}$, $\phi(\cdot; \Sigma)$ is the PDF of the normal vector with mean $\mathbf{0}$ and covariance Σ . The MLE of Σ maximizes the likelihood function defined as:

$$\begin{aligned} \ell(\Sigma; \mathbf{x}^i) &= \frac{1}{n} \sum_{i=1}^n \log \phi(\mathbf{f}^{-1}(\mathbf{x}^i); \Sigma) \\ &= c - \frac{1}{2} \log \det \Sigma - \frac{1}{2} \text{Tr} \left(\Sigma^{-1} \frac{1}{n} \sum_{i=1}^n \mathbf{z}^i (\mathbf{z}^i)^\top \right), \end{aligned} \quad (4)$$

over $\Sigma \in \mathcal{E}$, where $\mathbf{z}^i = \mathbf{f}^{-1}(\mathbf{x}^i)$ and c is a universal constant (We omit here and later the constant arising from $\left| \frac{\partial \mathbf{z}}{\partial \mathbf{x}} \right|$ after the log transformation). Thus the MLE of Σ is the sample covariance of $\mathbf{Z} := \mathbf{f}(\mathbf{X}) = [f_1(\mathbf{X}_1), \dots, f_p(\mathbf{X}_p)]$. When we substitute $\mathbf{f}$ by its empirical estimation in Eq. (2), the resulting covariance matrix $\hat{\Sigma}$ of $\hat{\mathbf{Z}} := \hat{\mathbf{f}}(\mathbf{X})$ is still consistent and asymptotically normal under some regularity conditions [30], which justifies the use of our estimator $\hat{\mathbf{f}}$. To simplify notation, we assume $\mathbf{f}$ is known below.

For a Gaussian copula, notice Σ is a correlation matrix, thus we update $\hat{\Sigma} = P_{\mathcal{E}} \hat{\Sigma}$, where $P_{\mathcal{E}}$ scales its argument to output a correlation matrix: for $D = \text{diag}(\Sigma)$, $P_{\mathcal{E}}(\Sigma) = D^{-1/2} \Sigma D^{-1/2}$. The obtained $\hat{\Sigma}$ is still consistent and asymptotically normal.

6.2 Incomplete Mixed Observations

When some columns are ordinal and some data is missing, the Gaussian latent vector $\mathbf{z}^i$ is no longer fully observed. We can compute the entries of $\mathbf{z}^i$ corresponding to continuous data: $\mathbf{z}_{C_i}^i = \mathbf{f}_{C_i}^{-1}(\mathbf{x}_{C_i}^i)$. However, for ordinal data, $\mathbf{f}_{D_i}^{-1}(\mathbf{x}_{D_i}^i)$ is a Cartesian product of intervals; we only know that $\mathbf{z}_{D_i}^i \in \mathbf{f}_{D_i}^{-1}(\mathbf{x}_{D_i}^i)$. The entries corresponding to missing observations, $\mathbf{z}_{M_i}^i$, are entirely unconstrained. Hence the latent matrix $\hat{\mathbf{Z}}$ is only incompletely observed, and it is no longer possible to simply compute its covariance.

We propose an expectation maximization (EM) algorithm to estimate Σ for incomplete mixed observation. Proceeding in an iterative fashion, we replace unknown $\mathbf{z}^i(\mathbf{z}^i)^\top$ with their expectation conditional on observations $\mathbf{x}_{O_i}^i$ and an estimate $\hat{\Sigma}$ in the E-step, then in the M-step we update the estimate of Σ as the conditional expectation of covariance matrix:

$$G(\hat{\Sigma}, \mathbf{x}_{O_i}^i) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\mathbf{z}^i(\mathbf{z}^i)^\top | \mathbf{x}_{O_i}^i, \hat{\Sigma}]. \quad (5)$$

Similar to the case of complete continuous data, we further scale the estimate to a correlation matrix. We first present the EM algorithm in Algorithm 2, then provide precise statements in Section 6.3. Computation details of Algorithm 2 appear in Section 6.4 and Section 6.5.

Algorithm 2 EM algorithm for Gaussian Copula

Input: observed entries $\{\mathbf{x}_{O_i}^i\}_{i=1}^n$.

Initialize: $t = 0, \Sigma^{(0)}$.

For $t = 0, 1, 2, \dots$

(1) E-step: Compute $G^{(t)} = G(\Sigma^{(t)}, \mathbf{x}_{O_i}^i)$.

(2) M-step: $\Sigma^{(t+1)} = G^{(t)}$.

(3) Scale to correlation matrix: $\Sigma^{(t+1)} = P_{\mathcal{E}}(\Sigma^{(t+1)})$

until convergence.

Output: $\hat{\Sigma} = \Sigma^{(t)}$.

6.3 EM algorithm

We first write down the marginal density of observed values by integrating out the missing data. Since $\mathbf{x}^i \sim \text{GC}(\Sigma, \mathbf{f})$, there exist latent $\mathbf{z}^i$ satisfying $\mathbf{f}(\mathbf{z}^i) = \mathbf{x}^i$ and $\mathbf{z}^i \sim \mathcal{N}_p(\mathbf{0}, \Sigma)$. The likelihood of Σ given observation $\mathbf{x}_{O_i}^i$ is the integral over the latent Gaussian vector $\mathbf{z}_{O_i}^i$ that maps to $\mathbf{x}_{O_i}^i$ under the marginal $\mathbf{f}_{O_i}$. Hence the observed log likelihood we seek to maximize is:

$$\ell_{\text{obs}}(\Sigma; \mathbf{x}_{O_i}^i) = \frac{1}{n} \sum_{i=1}^n \int_{\mathbf{z}_{O_i}^i \in \mathbf{f}_{O_i}^{-1}(\mathbf{x}_{O_i}^i)} \phi(\mathbf{z}_{O_i}^i; \mathbf{0}, \Sigma_{O_i, O_i}) d\mathbf{z}_{O_i}^i, \quad (6)$$

where Σ_{O_i, O_i} denote the submatrix of Σ with rows and columns in O_i . With known $\mathbf{f}$, MAR mechanism guarantees the maximizer of the likelihood in Eq. (6) shares the consistency and asymptotic normality of standard maximum likelihood estimate, according to the classical theory [18, Chapter 6.2].

However, the maximizer has no closed form expression. Even direct evaluation of $\ell_{\text{obs}}(\Sigma; \mathbf{x}_{O_i}^i)$ is challenging since it involves

multivariate Gaussian integrals in a truncated region and the observed locations O_i varies for different observations i . Instead, the proposed EM algorithm is guaranteed to monotonically converge to a local maximizer according to classical EM theory [21, Chapter 3].

Now we derive the proposed EM algorithm in detail. Suppose we know the values of the unobserved $\mathbf{z}^i$. Then the joint likelihood function is the same as in Eq. (4). Since the values of $\mathbf{z}^i$ are unknown, we treat $\mathbf{z}^i$ as latent variables and $\mathbf{x}_{O_i}^i$ as observed variables. Substituting the joint likelihood function by its expected value given observations $\mathbf{x}_{O_i}^i$ and an estimate $\hat{\Sigma}$:

$$\begin{aligned} Q(\Sigma; \hat{\Sigma}, \mathbf{x}_{O_i}^i) &:= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\ell(\Sigma; \mathbf{x}_{O_i}^i, \mathbf{z}^i) | \mathbf{x}_{O_i}^i, \hat{\Sigma}] \\ &= c - \frac{1}{2} \left(\log \det(\Sigma) + \text{Tr} \left(\Sigma^{-1} G(\hat{\Sigma}, \mathbf{x}_{O_i}^i) \right) \right). \end{aligned}$$

EM theory [21, Chapter 3] guarantees the updated $\tilde{\Sigma} = \text{argmax}_{\Sigma \in \mathcal{E}} Q(\Sigma; \hat{\Sigma}, \mathbf{x}_{O_i}^i)$ improves the likelihood with $\hat{\Sigma}$,

$$\ell_{\text{obs}}(\tilde{\Sigma}; \mathbf{x}_{O_i}^i) \geq \ell_{\text{obs}}(\hat{\Sigma}; \mathbf{x}_{O_i}^i),$$

and that by iterating this update, we produce a sequence $\{\Sigma^{(t)}\}$ that converges monotonically to a local maximizer of $\ell_{\text{obs}}(\Sigma; \mathbf{x}_{O_i}^i)$. At the t -th iteration, for the E step we compute $\mathbb{E}[\mathbf{z}^i(\mathbf{z}^i)^\top | \mathbf{x}_{O_i}^i, \Sigma^{(t)}]$ to express $Q(\Sigma; \Sigma^{(t)}, \mathbf{x}_{O_i}^i)$ in terms of Σ . For the M step, we find $\Sigma^{(t+1)} = \text{argmax}_{\Sigma} Q(\Sigma; \Sigma^{(t)}, \mathbf{x}_{O_i}^i)$. In practice, we resort to an approximation, as in [11]. Notice that the unconstrained maximizer is $\tilde{\Sigma} = G(\Sigma^{(t)}, \mathbf{x}_{O_i}^i)$. We update $\Sigma^{(t+1)} = P_{\mathcal{E}} \tilde{\Sigma}$.

6.4 Conditional Expectation Computation

Suppressing index i , we now show how to compute $\mathbb{E}[\mathbf{z}\mathbf{z}^\top | \mathbf{x}_O, \Sigma]$ in Eq. (5). With $\mathbf{z}_C = \mathbf{f}_C^{-1}(\mathbf{x}_C)$, it suffices to compute the following terms:

- (1) the conditional mean and covariance of observed ordinal dimensions $\mathbb{E}[\mathbf{z}_D | \mathbf{x}_O, \Sigma], \text{Cov}[\mathbf{z}_D | \mathbf{x}_O, \Sigma]$.
- (2) the conditional mean and covariance of missing dimensions $\mathbb{E}[\mathbf{z}_M | \mathbf{x}_O, \Sigma], \text{Cov}[\mathbf{z}_M | \mathbf{x}_O, \Sigma]$.
- (3) the conditional covariance between missing and observed ordinal dimensions $\text{Cov}[\mathbf{z}_M, \mathbf{z}_D | \mathbf{x}_O, \Sigma]$.

We show that with the results from (1), we can compute (2) and (3). Computation for (1) is put in Sec 6.5.

Suppose we can know the ordinal values $\mathbf{z}_D$ and thus $\mathbf{z}_O$. Conditional on $\mathbf{z}_O$, the missing dimensions $\mathbf{z}_M$ follows normal distribution with mean $\mathbb{E}[\mathbf{z}_M | \mathbf{z}_O, \Sigma] = \Sigma_{M,O} \Sigma_{O,O}^{-1} \mathbf{z}_O$. Further taking expectation of $\mathbf{z}_O$ conditional on observation, we obtain

$$\mathbb{E}[\mathbf{z}_M | \mathbf{x}_O, \Sigma] = \mathbb{E}[\mathbb{E}[\mathbf{z}_M | \mathbf{z}_O, \Sigma] | \mathbf{x}_O, \Sigma] = \Sigma_{M,O} \Sigma_{O,O}^{-1} \mathbb{E}[\mathbf{z}_O | \mathbf{x}_O, \Sigma].$$

One can compute $\text{Cov}[\mathbf{z}_M | \mathbf{x}_O, \Sigma]$ and $\text{Cov}[\mathbf{z}_M, \mathbf{z}_D | \mathbf{x}_O, \Sigma]$ similarly: deferring details to the supplement, we find

$$\text{Cov}[\mathbf{z}_M, \mathbf{z}_O | \mathbf{x}_O, \Sigma] = \Sigma_{M,O} \Sigma_{O,O}^{-1} \text{Cov}[\mathbf{z}_O | \mathbf{x}_O, \Sigma],$$

$$\begin{aligned} \text{Cov}[\mathbf{z}_M | \mathbf{x}_O, \Sigma] &= \Sigma_{M,M} - \Sigma_{M,O} \Sigma_{O,O}^{-1} \Sigma_{O,M} \\ &\quad + \Sigma_{M,O} \Sigma_{O,O}^{-1} \text{Cov}[\mathbf{z}_O | \mathbf{x}_O, \Sigma] \Sigma_{O,O}^{-1} \Sigma_{O,M}, \end{aligned}$$

where $\text{Cov}[\mathbf{z}_O|\mathbf{x}_O, \Sigma]$ has $\text{Cov}[\mathbf{z}_D|\mathbf{x}_O, \Sigma]$ as its submatrix and 0 elsewhere, $\text{Cov}[\mathbf{z}_M, \mathbf{z}_O|\mathbf{x}_O, \Sigma]$ has $\text{Cov}[\mathbf{z}_M, \mathbf{z}_D|\mathbf{x}_O, \Sigma]$ as its submatrix and 0 elsewhere.

6.5 Approximating Truncated Normal Moments

Now it remains to compute $E[\mathbf{z}_D|\mathbf{x}_O, \Sigma]$ and $\text{Cov}[\mathbf{z}_D|\mathbf{x}_O, \Sigma]$, which are the mean and covariance of a $|\mathcal{D}|$ -dimensional normal truncated to $\mathbf{f}_{\mathcal{D}}^{-1}(\mathbf{x}_D)$, a Cartesian product of intervals. The computation involves multiple integrals of a nonlinear function and only admits a closed form expression when $|\mathcal{D}| = 1$. Direct computational methods [4] are very expensive and can be inaccurate even for moderate $|\mathcal{D}|$. Notice the computation needs to be done for each row $\mathbf{x}_{O_i}^i$ at each EM iteration separately, thus sampling truncated normal distribution to evaluate the empirical moments [23] is still expensive for large number of data points n . Instead, we use a fast iterative method that scales well to large datasets, following [11].

Suppose all but one element of $\mathbf{z}_D$ is known. Then we can easily compute the resulting one dimensional truncated normal mean: for $j \in \mathcal{D}$, if z_j is unknown and $\mathbf{z}_{D-j}$ is known, let $E[z_j|\mathbf{z}_{D-j}, \mathbf{x}_O, \Sigma] := g_j(\mathbf{z}_{D-j}; \mathbf{x}_j, \Sigma)$ define the nonlinear function $g_j : \mathbb{R}^{|\mathcal{D}|-1} \rightarrow \mathbb{R}$, parameterized by $\mathbf{x}_j$ and Σ , detailed in the supplement. We may also use g_j to estimate $E[z_j|\mathbf{x}_O, \Sigma]$ if $E[\mathbf{z}_{D-j}|\mathbf{x}_O, \Sigma]$ is known:

$$\begin{aligned} E[z_j|\mathbf{x}_O, \Sigma] &= E[E[z_j|\mathbf{z}_{D-j}, \mathbf{x}_O, \Sigma]|\mathbf{x}_O, \Sigma] \\ &= E[g_j(\mathbf{z}_{D-j}; \mathbf{x}_j, \Sigma)|\mathbf{x}_O, \Sigma] \approx g_j(E[\mathbf{z}_{D-j}|\mathbf{x}_O, \Sigma]; \mathbf{x}_j, \Sigma), \end{aligned} \quad (7)$$

if g_j is approximately linear. In other words, we can iteratively update the marginal mean of $E[\mathbf{z}_D|\mathbf{x}_O, \Sigma]$. At EM iteration $t + 1$, we conduct one iteration update with initial value from last EM iteration $\hat{\mathbf{z}}_D^{(t)} \approx E[\mathbf{z}_D|\mathbf{x}_O, \Sigma^{(t)}]$:

$$E[z_j|\mathbf{x}_O, \Sigma^{(t+1)}] \approx \hat{z}_j^{(t+1)} := g_j(\hat{\mathbf{z}}_{D-j}^{(t)}; \mathbf{x}_j, \Sigma^{(t+1)}). \quad (8)$$

Surprisingly, one iteration update works well and more iterations do not bring significant improvement.

We use a diagonal approximation for $\text{Cov}[\mathbf{z}_D|\mathbf{x}_O, \Sigma]$: we approximate $\text{Cov}[z_j, z_k|\mathbf{x}_O, \Sigma]$ as 0 for $j \neq k \in \mathcal{D}$. This approximation performs well when z_j and z_k are nearly independent given all observed information. We approximate the diagonal entries $\text{Var}[z_j|\mathbf{x}_O, \Sigma^{(t+1)}]$ for $j \in \mathcal{D}$ using a recursion similar to Eq. (8), detailed in the supplement.

We point out the estimated covariance matrix in Eq. (5) is the sum of the sample covariance matrix of the imputed $\mathbf{z}^i$ using its conditional mean and the expected covariance brought by the imputation. The diagonal approximation only applies to the second term, while the first term is dense. Consequently, the estimator in Eq. (5) is dense and can fit a large range of covariance matrices. Empirical evidence indicates that our approximation even outperforms the MCMC algorithm without such diagonal approximation [14], shown in Section 7.1.

6.6 Computation Cost

The complexity of each EM iteration is $O(\alpha np^3)$ with observed entry ratio α . The overall complexity is $O(T\alpha np^3)$, where T is the number of EM steps required for convergence. We found $T \leq 50$ in most of our experiments. On a laptop with Intel-i5-3.1GHz Core

and 8 GB RAM, it takes 1.2min for our algorithm to converge on a dataset with size 2000×60 and 25% missing entries (generated as in Section 7.1 when $p = 60$). Scaling our algorithm to large p is important future work. However, our algorithm is usually faster than many start-of-the-art imputation algorithms for large n small p . Speed comparison on a dataset with size 6039×207 is shown in Section 7.3.

7 EXPERIMENTS

Our first experiment demonstrates that our method, Copula-EM, is able to estimate a well-specified Gaussian copula model faster than the MCMC method sbgcp [13, 14]. Our other experiments compare the accuracy of imputations produced by Copula-EM with missForest [29], xPCA [1] and imputeFAMD [2], state-of-the-art nonparametric imputation algorithms for mixed data; and the low rank matrix completion algorithms softImpute [20] and GLRM [32], which scale to large datasets. missForest is implemented with recommended default settings: 10 maximum iterations and 100 trees [28]. All other methods require selecting either the rank or the penalization parameter. We select them through 5-fold cross validation (5CV), unless otherwise specified. See the supplement for implementation details. For real datasets, we report results from our Copula-EM but put that from sbgcp in the supplement, since Copula-EM outperforms on all evaluation metrics and converges substantially faster.

To measure the imputation error on columns in I , we define a scaled mean absolute error (SMAE):

$$\text{SMAE} := \frac{1}{|I|} \sum_{j \in I} \frac{\|\hat{\mathbf{X}}_j - \mathbf{X}_j\|_1}{\|\mathbf{X}_j^{\text{med}} - \mathbf{X}_j\|_1},$$

where $\hat{\mathbf{X}}_j, \mathbf{X}_j^{\text{med}}$ are the imputed values and observed median for j -th column, respectively. The estimator's SMAE is smaller than 1 if it outperforms column median imputation. For each data type, the SMAE can be computed on corresponding columns. To evaluate the estimated correlation, we use relative error $\|\hat{\Sigma} - \Sigma\|_F / \|\Sigma\|_F$, where $\hat{\Sigma}$ is the estimated correlation matrix.

7.1 Synthetic Data

The first experiment compares the speed of the two algorithms to estimate Gaussian copula models: Copula-EM and sbgcp. Note Copula-EM is implemented in pure R, while the computational core of sbgcp is implemented in C. Hence further acceleration of Copula-EM is possible.

We generate 100 synthetic datasets with $n = 2000$ observations and $p = 15$ variables from a well-specified Gaussian copula model with random Σ generated [24]. For each Σ , first generate rows of $\mathbf{Z} \in \mathbb{R}^{n \times p}$ as $\mathbf{z}^1, \dots, \mathbf{z}^n \stackrel{i.i.d.}{\sim} \mathcal{N}(\mathbf{0}, \Sigma)$. Then generate $\mathbf{X} = \mathbf{f}(\mathbf{Z})$ using monotone $\mathbf{f}$ such that $\mathbf{X}_1, \dots, \mathbf{X}_5$ have exponential distributions, $\mathbf{X}_6, \dots, \mathbf{X}_{10}$ are binary and $\mathbf{X}_{11}, \dots, \mathbf{X}_{15}$ are 1-5 ordinal.

We randomly remove 30% of the entries of $\mathbf{X}$, train Copula-EM and sbgcp, and compute the imputation error on the held-out set. We plot the imputation accuracy and correlation estimation accuracy versus runtime of each algorithm in Figure 4. Copula-EM converges quickly, in about 25s, while sbgcp takes much longer

and suffers high error at shorter times. Copula-EM estimates correlations and continuous imputations at convergence more accurately than sbgscop even when the latter algorithm is given 6 times more runtime. Interestingly, Copula-EM recovers the correlation matrix better than sbgscop even asymptotically. These results demonstrate the impact of the approximate EM algorithm 6.5 compared to the (fully accurate) MCMC model of sbgscop: the approximation allows faster convergence, to an estimate of nearly the same quality.

For ordinal data imputation, Copula-EM reaches the same performance as sbgscop 6 times faster. For binary data imputation, sbgscop is four times slower than Copula-EM at reaching the final performance of Copula-EM, but sbgscop outperforms Copula-EM given even more time. We conjecture that the drop in imputation accuracy of Copula-EM for binary data could be mitigated using multiple imputation [18, Chapter 5.4], as outlined in Sec 4 by combining the imputations (using mean or median) into a single imputation to reduce the effect of approximating the truncated normal distribution.

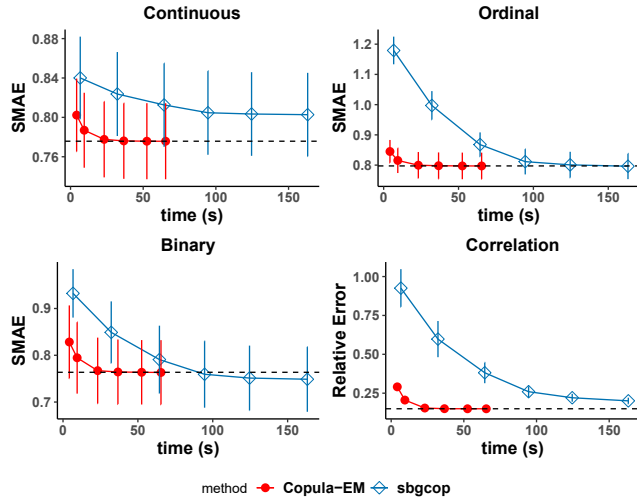


Figure 4: Copula-EM vs sbgscop: The imputation error for each data type and estimated correlation error over time cost. Dashed line indicates the final error of Copula-EM.

The second experiment compares the imputation accuracy of Copula-EM and nonparametric algorithms. Using the same data generation mechanism, we randomly remove 10% – 50% of the entries of X . The optimal rank selected using 5CV is 3 for xPCA and 6 for imputeFAMD. Shown in Figure 5, Copula-EM substantially outperforms all nonparametric algorithms for all data types.

7.2 General Social Survey (GSS) Data

We chose 18 variables with 2538 observations from GSS dataset in year 2014. 24.9% of the entries are missing. The dataset consists of 1 continuous (AGE) and 17 ordinal variables with 2 to 48 levels. We investigate the imputation accuracy on five selected variables: INCOME, LIFE, HEALTH, CLASS¹ and HAPPY. For each variable, we sample 1500 observation and divide them into 20 folds. We mask

¹Subjective class identification from lower to upper class

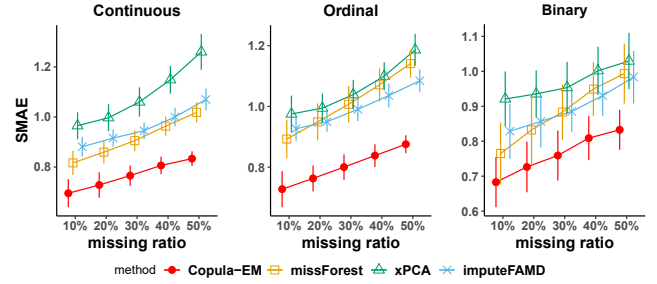


Figure 5: Copula-EM vs nonparametric algorithms: The imputation error for each data type on synthetic data.

Table 1: Imputation Error on Five GSS Variables

Variable	Copula-EM	missForest	xPCA	imputeFAMD
CLASS	0.735(0.10)	0.782(0.09)	0.795(0.08)	0.797(0.10)
LIFE	0.759(0.12)	0.828(0.17)	0.783(0.11)	0.821(0.11)
HEALTH	0.877(0.09)	1.143(0.18)	0.908(0.10)	0.947(0.04)
HAPPY	0.896(0.08)	1.079(0.15)	1.003(0.15)	1.001(0.10)
INCOME	0.869(0.07)	0.944(0.18)	1.090(0.15)	0.996(0.01)

one fold of only one variable as test data in each experiment. The selected rank is 2 for both xPCA and imputeFAMD. We report the SMAE for each variable in Table 1. Our method performs the best for all variables. Further our method always performs better than median imputation. In contrast, the other three methods perform worse than median imputation for some variables. Our method also provides estimated variable correlation, which is usually desired in social survey study. We plot high correlations from the copula correlation matrix as a graph in Figure 6.

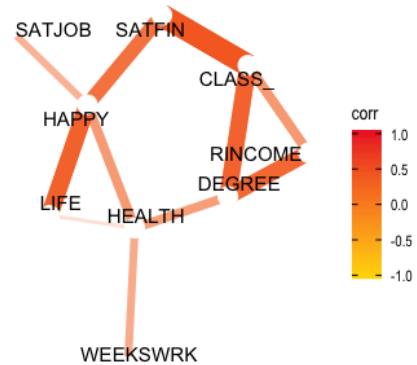


Figure 6: High Correlations ($|\cdot| > 0.3$) of 5 interesting variables from GSS data are plotted.

7.3 MovieLens 1M Data

Recall our method scales cubically in the number of variables. Hence for this experiment, we sample the subset of the MovieLens 1M data [12] consisting of the 207 movies with at least 1000 ratings

Table 2: Imputation Error on 207 Movies

Algorithm	MAE	RMSE
Column Median	0.702(0.004)	1.001(0.004)
Copula-EM	0.579(0.004)	0.880(0.005)
GLRM	0.595(0.004)	0.892(0.004)
softImpute	0.602(0.004)	0.883(0.004)
xPCA	0.613(0.004)	0.897(0.004)
missForest	0.669(0.004)	1.015(0.006)

and all users who rate at least one of those 207 movies. On this subset, 75.6% of entries are missing. Under the time limit 1 hour, we implement all algorithms but imputeFAMD. Copula-EM takes 9 mins and missForest takes 25 mins. These two methods have no parameters to tune. To select tuning parameters for other algorithms, we manually mask 10% of the data for the test set and use the remaining data to train the model, and repeat 20 times. The selected rank using 5CV is 99 for softImpute, 6 for xPCA and 8 for GLRM with bigger-vs-smaller loss. With the selected tuning parameter, low rank matrix completion methods are substantially faster. For example, softImpute only takes 33s. However, counting the additional time to select tuning parameters using 5CV, softImpute takes 16mins to select the penalization parameter with regularization path length 50, which is already more expensive than Copula-EM. Interestingly, the ranks selected are quite different even when the models perform similarly: GLRM chooses rank 8 while softImpute chooses rank 99.

We report both mean absolute error (MAE) and RMSE in Table 2. Our method outperforms all others in both MAE and RMSE. This result is notable, because Copula-EM does not directly minimize MAE or RMSE, while softImpute directly minimizes RMSE. It also indicates Copula-EM does not overfit even with $O(p^2)$ free parameters.

7.4 Music Auto-tagging: CAL500exp Data

The CAL500 expansion (CAL500exp) dataset [36] is an enriched version of the well-known CAL500 dataset [31]. This dataset consists of 67 binary tags (including genre, mood and instrument, labeled by experts) to 3223 music fragments from 500 songs. Music auto-tagging is a multi-label learning problem. A feature vector is usually computed first based on the music files and then a classifier is trained for each tag. This procedure is expensive and neglects the association among known labels. We treat this task as a missing data imputation problem and only use observed labels to impute unknown labels. This dataset is completely observed. We randomly remove some portions of the observed labels as a test set and repeat 20 times. The selected optimal rank is 4 for xPCA and 15 for imputeFAMD. Shown in Table 3, Copula-EM performs the best in terms of SMAE. The superiority of Copula-EM over other algorithms substantially grows as the missing ratio increases. Moreover, Copula-EM yields very stable imputations: the standard deviation of its SMAE is imperceptibly small.

Table 3: Imputation Error (SMAE) on CAL500exp.

Algorithm	40% missing	50% missing	60% missing
Copula-EM	0.799(0.002)	0.822(0.003)	0.849(0.002)
missForest	0.800(0.018)	0.984(0.026)	1.181(0.024)
imputeFAMD	0.823(0.013)	0.920(0.016)	1.114(0.020)
xPCA	0.911(0.018)	0.988(0.071)	1.108(0.145)

7.5 More Ordinal Data and Mixed Data

We compare mixed data imputation algorithms on two more ordinal classification datasets², Lecturers Evaluation (*LEV*) and Employee Selection (*ESL*), and two more mixed datasets, German Breast Cancer Study Group (*GBSG*)³ and Restaurant Tips (*TIPS*)⁴. Dataset descriptions appear in Table 4, and more details appear in the supplement. All datasets are completely observed.

For each dataset, we randomly remove 30% entries as a test set and repeat 100 times. For ordinal classification datasets, we evaluate the SMAE for the label and for the features, respectively. For mixed datasets, we evaluate the SMAE for ordinal dimensions and for continuous dimensions, respectively. We report results in Table 4. Our method outperforms the others in all but one setting, often by a substantial margin.

8 SUMMARY AND DISCUSSION

In this paper, we proposed an imputation algorithm that models mixed data with a Gaussian copula model, together with an effective approximate EM algorithm to estimate the copula correlation with incomplete mixed data. Our algorithm has no tuning parameter and are easy to implement. Our experiments demonstrate the success of the proposed method. Scaling these methods to larger datasets (especially, with more columns), constitutes important future work.

We end by noting a few contrasts between the present approach and typical low rank approximation methods for data imputation. Low rank approximation constructs a latent simple (low rank) object and posits that observations are noisy draws from that simple latent object. In contrast, our approach uses a parametric, but full-dimensional, model for the latent object; observations are given by a deterministic function of the latent object. In other words, in previous work the latent object is exact and the observations are noisy; in our work, the latent object is noisy and the observations are exact. Which more faithfully models real data? As evidence, we might consider whether low rank models agree on the best rank to fit a given dataset. For example, on the MovieLens dataset: (1) The low rank matrix completion methods xPCA and GLRM, implemented using alternating minimization, select small optimal ranks (6 and 8), while softImpute, implemented using nuclear norm minimization, selects the much larger optimal rank 99. (2) Our algorithm outperforms all the low rank matrix completion methods we tested. These observations suggest the low rank assumption commonly used to fit the MovieLens dataset may not be fundamental, but may arise as a mathematical artifact [33]. More supporting empirical

² Available at <https://waikato.github.io/weka-wiki/datasets/>

³ Available at <https://cran.r-project.org/web/packages/mfip/>

⁴ Available at <http://ggobi.org/book/>

Table 4: Imputation Error on More Ordinal and Mixed Datasets.

Dataset	Size	Selected Rank	Type	Copula-EM	missForest	xPCA	imputeFAMD
ESL	488 × 5 4 features, 1 label	1 (xPCA)	Label	0.372(0.04)	0.553(0.08)	0.404(0.04)	0.503(0.06)
		5 (imputeFAMD)	Feature	0.584(0.03)	0.873(0.06)	0.668(0.03)	0.687(0.03)
LEV	1000 × 5 4 features, 1 label	1 (xPCA)	Label	0.750(0.04)	0.970(0.09)	0.860(0.06)	0.882(0.05)
		5 (imputeFAMD)	Feature	0.907(0.01)	0.799(0.03)	1.037(0.02)	1.085(0.04)
GBSG	686 × 10 6 continuous, 4 ordinal	2 (xPCA)	Ordinal	0.793(0.03)	0.887(0.05)	0.876(0.04)	0.840(0.03)
		2 (imputeFAMD)	Continuous	0.876(0.01)	1.029(0.03)	1.100(0.04)	1.038(0.03)
TIPS	244 × 7 2 continuous, 5 ordinal	2 (xPCA)	Ordinal	0.786(0.05)	0.928(0.09)	0.928(0.08)	0.891(0.09)
		6 (imputeFAMD)	Continuous	0.755(0.04)	0.837(0.05)	1.011(0.11)	0.892(0.13)

results can be found in [3]: the performance of softImpute keeps improving as the rank increases (up to 10^3).

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A COMPUTATIONAL DETAILS

A.1 Details for Section 6.5

Denote the observation $\{\mathbf{x}_O, \Sigma\}$ i.e. $\{\mathbf{z}_C = \mathbf{f}_C^{-1}(\mathbf{x}_C), \mathbf{z}_D \in \mathbf{f}_D^{-1}(\mathbf{x}_D), \Sigma\}$ as $\{*\}$. Since the task is to compute the marginal mean and variance of a multivariate truncated normal, we suppose $\mathcal{M} = \emptyset$ here without loss of generality. For each $j \in \mathcal{D}$, we use the law of total expectation by conditioning on $\mathbf{z}_{D-j}$ first. Given $\{*, \mathbf{z}_{D-j}\}$, $\mathbf{z}_j$ is univariate normal with mean $\tilde{\mu}_j = \Sigma_{j,-j}\Sigma_{-j,-j}^{-1}\mathbf{z}_{-j}$ and variance $\tilde{\sigma}_j^2 = 1 - \Sigma_{j,-j}\Sigma_{-j,-j}^{-1}\Sigma_{-j,j}$, truncated to the region $f_j^{-1}(x_j)$, where the index $-j$ means all dimensions but j , i.e., $[p] \setminus j$. The region $f_j^{-1}(x_j)$ is an interval: $f_j^{-1}(x_j) = (a_j, b_j]$. Here are three cases: (1) $a_j, b_j \in \mathbb{R}$; (2) $a_j \in \mathbb{R}, b_j = \infty$; (3) $a_j = -\infty, b_j \in \mathbb{R}$. The computation for all cases are similar. We take the first case as an example. First we introduce a lemma describing the first and second moments of a truncated univariate normal.

Lemma 5. Suppose a univariate random variable $z \sim \mathcal{N}(\mu, \sigma^2)$. For constants $a < b$, let $\alpha = (a - \mu)/\sigma$ and $\beta = (b - \mu)/\sigma$. Then the mean and variance of z truncated to the interval $(a, b]$ are:

$$\mathbb{E}(z|a < z \leq b) = \mu + \frac{\phi(\alpha) - \phi(\beta)}{\Phi(\beta) - \Phi(\alpha)} \cdot \sigma$$

$$\text{Var}(z|a < z \leq b) = \left(1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{\Phi(\beta) - \Phi(\alpha)} - \left(\frac{\phi(\alpha) - \phi(\beta)}{\Phi(\beta) - \Phi(\alpha)}\right)^2\right) \sigma^2.$$

Plugging $\mu = \tilde{\mu}_j, \sigma^2 = \tilde{\sigma}_j^2$ and $(a, b] = f_j^{-1}(x_j)$ into the above mean and variance formulas, we obtain the expression of $g_j(\mathbf{z}_{D-j}; x_j, \Sigma)$ defined in Section 6.5, and the univariate truncated normal variance $\text{Var}[\mathbf{z}_j|\mathbf{z}_{D-j}, \mathbf{x}_O, \Sigma] =: h_j(\mathbf{z}_{D-j}; x_j, \Sigma)$, a nonlinear function $\mathbb{R}^{|\mathcal{D}|-1} \rightarrow \mathbb{R}$, parameterized by x_j and Σ . Write down the formula for marginal variance conditional on observation:

$$\begin{aligned} \text{Var}[\mathbf{z}_j|*] &= \mathbb{E}[\text{Var}[\mathbf{z}_j|\mathbf{z}_{D-j}, *]] + \text{Var}[\mathbb{E}[\mathbf{z}_j|\mathbf{z}_{D-j}, *]] \\ &= \mathbb{E}[h_j(\mathbf{z}_{D-j}; x_j, \Sigma)] + \text{Var}[g_j(\mathbf{z}_{D-j}; x_j, \Sigma)] \end{aligned}$$

We approximate the first term as $h_j(\mathbb{E}[\mathbf{z}_{D-j}|*]; x_j, \Sigma)$. As for the second term, Guo et al. [11] approximated it as $\text{Var}[\tilde{\mu}_j|*]$ based on $\mathbb{E}[g_j^2(\mathbf{z}_{D-j}; x_j, \Sigma)] \approx g_j^2(\mathbb{E}[\mathbf{z}_{D-j}|*]; x_j, \Sigma)$. However, we found in practice simply dropping the second term performs better.

In summary, given an estimate $\hat{\mathbf{z}}_D^{(t)} \approx \mathbb{E}[\mathbf{z}_D|\mathbf{x}_O, \Sigma^{(t)}]$ and $\Sigma^{(t+1)}$, for $j \in \mathcal{D}$, we update $\mathbb{E}[\mathbf{z}_j|\mathbf{x}_O, \Sigma^{(t+1)}] \approx g_j(\hat{\mathbf{z}}_{D-j}^{(t)}; x_j, \Sigma^{(t+1)})$ and $\text{Var}[\mathbf{z}_j|\mathbf{x}_O, \Sigma^{(t+1)}] \approx h_j(\hat{\mathbf{z}}_{D-j}^{(t)}; x_j, \Sigma^{(t+1)})$. In other words, we update the conditional mean and variance of $\mathbf{z}_j$ as the univariate truncated normal mean and variance with all other observed ordinal dimensions equal to their mean from last iteration, i.e. $\mathbf{z}_{D-j} = \hat{\mathbf{z}}_{D-j}^{(t)}$.

A.2 Details for Section 6.4

Given $\mathbb{E}[\mathbf{z}_O|*], \mathbb{E}[\mathbf{z}_M|*]$ and $\text{Cov}[\mathbf{z}_O|*]$, it suffices to compute $\mathbb{E}[\mathbf{z}_M\mathbf{z}_O^T|*]$ and $\mathbb{E}[\mathbf{z}_M\mathbf{z}_M^T|*]$ for $\text{Cov}[\mathbf{z}_M, \mathbf{z}_O|*]$ and $\text{Cov}[\mathbf{z}_M|*]$. Using the law of total expectation, we have:

$$\begin{aligned} \mathbb{E}[\mathbf{z}_M\mathbf{z}_O^T|*] &= \mathbb{E}[\mathbb{E}[\mathbf{z}_M\mathbf{z}_O^T|\mathbf{z}_O, *]] = \mathbb{E}[\mathbb{E}[\mathbf{z}_M|\mathbf{z}_O, *] \cdot \mathbf{z}_O^T|*] \\ &= \mathbb{E}[\Sigma_{M,O}\Sigma_{O,O}^{-1}\mathbf{z}_O \cdot \mathbf{z}_O^T|*] = \Sigma_{M,O}\Sigma_{O,O}^{-1}\mathbb{E}[\mathbf{z}_O\mathbf{z}_O^T|*]. \end{aligned}$$

$$\begin{aligned} \mathbb{E}[\mathbf{z}_M\mathbf{z}_M^T|*] &= \mathbb{E}[\mathbb{E}[\mathbf{z}_M\mathbf{z}_M^T|\mathbf{z}_O, *]] \\ &= \mathbb{E}[\text{Cov}[\mathbf{z}_M|\mathbf{z}_O, *]] + \mathbb{E}[\mathbb{E}[\mathbf{z}_M|\mathbf{z}_O, *] \cdot \mathbb{E}[\mathbf{z}_M^T|\mathbf{z}_O, *]] \\ &= \Sigma_{M,M} - \Sigma_{M,O}\Sigma_{O,O}^{-1}\Sigma_{O,M} + \Sigma_{M,O}\Sigma_{O,O}^{-1}\mathbb{E}[\mathbf{z}_O|*]\mathbb{E}[\mathbf{z}_O^T|*]\Sigma_{O,O}^{-1}\Sigma_{O,M}. \end{aligned}$$

B SUPPLEMENT FOR EXPERIMENTS

B.1 Implementation Details

For softImpute, we first center the rows and columns, then select the penalization parameter in the path from 45 (rank 12) to 6 (rank 207) with 50 points. For GLRM, we use quadratic regularization on X factor and ordinal regularization on Y factor. The model is fitted with SVD initialization and offset term. After a small grid search, we select the quadratic regularization parameter as $n_{\text{obs}} \times 1.2 \times 10^{-4}$ where n_{obs} is the number of observed entries. Then the rank is selected through an exhaustive search. For xPCA and imputeFAMD, the rank is selected through an exhaustive search.

B.2 Results of sbgcp on Real Datasets

For GSS data, Copula-EM takes 24s, while sbgcp with 1000 iterations takes 87s, with imputation error: CALSS, 0.992(0.13); LIFE, 0.924(0.7); HEALTH, 1.132(0.15); HAPPY, 1.231(0.11); INCOME, 0.931(0.03).

For movielens data, Copula-EM takes 9 mins, while sbgcp with 200 iterations takes 33 mins, with imputation error: MAE, 0.752(0.004); RMSE, 1.030(0.005).

For CAL500exp data, Copula-EM takes 80s, while sbgcp with 500 iterations takes 290s, with imputation error: 1.301(0.019) for 40% missing ratio; 1.328(0.015) for 50% missing ratio; 1.379(0.016) for 60% missing ratio.

For four small datasets used in Section 7.5, the time sbgcp with 1000 iterations takes is 2 times to 9 times (varying over datasets) of the time Copula-EM takes. The corresponding imputation error is: ESL label 0.466(0.04), feature 0.649(0.02); LEV label 0.849(0.03), feature 0.936(0.01); GBSG ordinal 0.992(0.03), continuous 0.953(0.02); TIPS ordinal 0.984(0.06), continuous 0.768(0.05).

B.3 Datasets Description for Section 7.5

ESL This dataset contains profiles of applicants for certain jobs.

The recruiting company, based upon psychometric test results and interviews with the candidates, determined the values of the input attributes. The output is an overall score corresponding to the degree of fitness of the candidate.

LEV This dataset contains lecturer evaluations. Students evaluate their lecturers according to four attributes such as oral skills and contribution to their professional/general knowledge. The output is an overall score of the lecturer's performance.

GBSG This dataset contains the information of women with breast cancer concerning the status of the tumours and the hormonal system of the patient.

TIPS This dataset concerns the tips given to a waiter in a restaurant collected from customers. Recording variables contains the price of the meal, the tip amount and the conditions of the restaurant meal (number of guests, time of data, etc.).

C PROOF OF LEMMAS

Proof of Lemma 1.

PROOF. For any $j \in [p]$, $x_j \stackrel{d}{=} f_j(z_j)$ if and only if (iff) x_j and $f_j(z_j)$ have the same CDF. For each $j \in [p]$, since f_j^{-1} exists for any strictly monotone f_j , we can calculate the CDF of $f_j(z_j)$:

$$F_{f_j(z_j)}(t) = P(f_j(z_j) \leq t) = P(z_j \leq f_j^{-1}(t)) = \Phi(f_j^{-1}(t)).$$

Then $x_j \stackrel{d}{=} f_j(z_j)$ iff $\Phi \circ f_j^{-1} = F_j$, equivalently, $f_j = F_j^{-1} \circ \Phi$. $\square$

Proof of Lemma 2.

PROOF. It suffices to show for monotone function f , $x \stackrel{d}{=} f(z)$ iff $f(z) = \text{cutoff}(z; S)$ with $S = \{s_l = F_z^{-1}(\sum_{t=1}^l p_t) : l \in [k-1]\}$.

Notice $x \stackrel{d}{=} f(z)$ iff the range of $f(z)$ is $[k]$ and $p_l = P(f(z) = l)$ for any $l \in [k]$. When $f(z) = \text{cutoff}(z; S)$, further define $s_k = \infty$ and $s_0 = -\infty$. Since z is continuous with CDF F_z , it suffices to show:

$$P(f(z) = l) = P(s_{l-1} < z \leq s_l) = F_z(s_l) - F_z(s_{l-1}) = p_l, \text{ for } l \in [k]$$

When $x \stackrel{d}{=} f(z)$, $f(z)$ has range $[k]$. For $l \in [k]$, define $A_l = \{z : f(z) = l\}$, $s_l = \sup_{z \in A_l} z$ and $s_0 = \inf_{z \in A_1} z$. Since $P(f(z) = l) = p_l > 0$, we have $\inf_{z \in A_l} z < s_l$. Since f is monotone, we have $s_{l-1} \leq \inf_{z \in A_l} z$.

Claim $s_{l-1} = \inf_{z \in A_l} z$. If not, there exists $s_{l-1} < z^* < \inf_{z \in A_l} z$ satisfying $(l-1) \leq f(z^*) \leq l$. Since $f(z)$ has range $[k]$, $f(z^*)$ can only be l or $l-1$. Equivalently $z^* \in A_l$ or $z^* \in A_{l-1}$, which contradicts $s_{l-1} < z^* < \inf_{z \in A_l} z$. Thus $s_{l-1} = \inf_{z \in A_l} z$, $f(z) = 1 + \sum_{l=1}^{k-1} \mathbb{1}(z > s_l)$,

$$p_l = P(f(z) = l) = P(z \in A_l) = P(s_{l-1} \leq z \leq s_l) = F_z(s_l) - F_z(s_{l-1}),$$

$$\text{Thus we have } F_z(s_l) = \sum_{t=1}^l p_t \Rightarrow s_l = F_z^{-1}(\sum_{t=1}^l p_t).$$

$\square$

Proof of Lemma 3. Before we prove Lemma 3, we introduce the Dvoretzky-Kiefer-Wolfowitz inequality [7], also introduced in [17].

THE DVORETZKY-KIEFER-WOLFOWITZ INEQUALITY. For any i.i.d. sample $x^1, \dots, x^n$ with distribution F , then when $\epsilon > 0$,

$$P\left(\sup_{t \in \mathbb{R}} |\mathbb{F}_n(t) - F(t)| \geq \epsilon\right) \leq 2e^{-2n\epsilon^2}, \text{ where } \mathbb{F}_n(t) = \frac{\sum_{i=1}^n \mathbb{1}\{x^i \leq t\}}{n}$$

PROOF. Applying the Dvoretzky-Kiefer-Wolfowitz inequality, for any $\epsilon > 0$, $\Pr(\sup_{t \in \mathbb{R}} |\mathbb{F}_n(t) - F(t)| < \epsilon) \geq 1 - 2e^{-2n\epsilon^2}$.

Take $\epsilon > n^{-1}$, $\sup_{t \in \mathbb{R}} \left|\frac{n}{n+1} \mathbb{F}_n(t) - F(t)\right| < 2\epsilon$. Further let $\epsilon < K_1 \triangleq \min\left\{\frac{F(m)}{4}, \frac{1-F(M)}{4}\right\}$, we have $\frac{n}{n+1} \mathbb{F}_n(t) \in [\frac{F(m)}{2}, \frac{1+F(M)}{2}]$ for $t \in [m, M]$. Then,

$$\begin{aligned} \sup_{t \in [m, M]} |\hat{f}^{-1}(t) - f^{-1}(t)| &= \sup_{t \in [m, M]} \left| \Phi^{-1}\left(\frac{n}{n+1} \mathbb{F}_n(t)\right) - \Phi^{-1}(F(t)) \right| \\ &\leq \sup_{r \in [\frac{F(m)}{2}, \frac{1+F(M)}{2}]} \left| \left(\Phi^{-1}(r)\right)' \right| \cdot \sup_{t \in [m, M]} \left| \frac{n}{n+1} \mathbb{F}_n(t) - F(t) \right| \\ &< 2\epsilon \cdot \sup_{r \in [\frac{F(m)}{2}, \frac{1+F(M)}{2}]} \left| \left(\Phi^{-1}(r)\right)' \right| \end{aligned}$$

Since $(\Phi^{-1}(r))' = \frac{1}{\phi(\Phi^{-1}(r))}$, we get $\sup_{r \in [\frac{F(m)}{2}, \frac{1+F(M)}{2}]} |(\Phi^{-1}(r))'| = K_2 \triangleq 1/\min\left\{\phi\left(\Phi^{-1}\left(\frac{F(m)}{2}\right)\right), \phi\left(\Phi^{-1}\left(\frac{F(M)+1}{2}\right)\right)\right\}$. Adjusting the constants, for $2K_2n^{-1} < \epsilon < 2K_1K_2$, we have

$$P\left(\sup_{t \in [m, M]} |\hat{f}^{-1}(t) - f^{-1}(t)| > \epsilon\right) \leq 2 \exp\left\{-\frac{n\epsilon^2}{2K_2^2}\right\}.$$

$\square$

Proof of Lemma 4. Before we prove Lemma 4, we introduce the Bretagnolle-Huber-Carol inequality introduced in [34].

THE BRETAGNOLLE-HUBER-CAROL INEQUALITY. If the random vector $(N_1, \dots, N_k)$ is multinomially distributed with parameters n and $(p_1, \dots, p_k)$, then

$$P\left(\sum_{i=1}^k |N_i/n - p_i| \geq \epsilon\right) \leq 2^k e^{-\frac{1}{2}n\epsilon^2}, \quad \epsilon > 0.$$

PROOF. According to Lemma 2, the cutoff function $f(z) = \text{cutoff}(z; S)$ is unique and $S = \{s_l : s_l = \Phi^{-1}(\sum_{t=1}^l p_t), l \in [k-1]\}$. Define $s_l^* = \Phi^{-1}\left(\frac{\sum_{i=1}^n \mathbb{1}(x^i \leq l)}{n}\right)$ for $l \in [k-1]$, $s_0^* = -\infty$, $s_k^* = \infty$, and $\Delta_l^* = \Phi(s_l^*) - \Phi(s_{l-1}^*) = \sum_{i=1}^n \mathbb{1}(x^i = l)/n$. Notice $(n\Delta_1^*, \dots, n\Delta_k^*)$ is multinomially distributed with parameters n and $(p_1, \dots, p_k)$, applying the Bretagnolle-Huber-Carol inequality, for any $\epsilon > 0$, with probability at least $1 - 2^k e^{-\frac{1}{2}n\epsilon^2}$, $\sum_{l=1}^k |\Delta_l^* - p_l| < \epsilon$. First for each $l \in [k]$, $|\Phi(s_l^*) - \Phi(s_l)| \leq \sum_{t=1}^k |\Delta_t^* - p_t| < \epsilon$. Take $\epsilon > n^{-1}$, we have

$$\left|\Phi(s_l^*) \cdot \frac{n}{n+1} - \Phi(s_l)\right| \leq |\Phi(s_l^*) - \Phi(s_l)| + \frac{\Phi(s_l^*)}{n+1} < 2\epsilon$$

$$\Phi(s_l) - 2\epsilon < \Phi(s_l^*) \cdot \frac{n}{n+1} = \frac{\sum_{i=1}^n \mathbb{1}(x^i \leq l)}{n+1} < \Phi(s_l) + 2\epsilon$$

When $l \in [k-1]$, we have $p_1 \leq \Phi(s_l) \leq \sum_{t=1}^{k-1} p_t$. Further let $\epsilon < K_1 \triangleq \min\left\{\frac{p_1}{4}, \frac{p_k}{4}\right\}$, we have $\frac{p_1}{2} \leq \Phi(s_l^*) \cdot \frac{n}{n+1} \leq 1 - \frac{p_k}{2}$. Thus:

$$\begin{aligned} \|\hat{S} - S\|_1 &= \sum_{l=1}^{k-1} |\hat{s}_l - s_l| = \sum_{l=1}^{k-1} \left| \Phi^{-1}\left(\frac{\sum_{i=1}^n \mathbb{1}(x^i \leq l)}{n+1}\right) - \Phi^{-1}(\Phi(s_l)) \right| \\ &\leq \sup_{r \in [\frac{p_1}{2}, 1 - \frac{p_k}{2}]} \left| \left(\Phi^{-1}(r)\right)' \right| \cdot \sum_{l=1}^{k-1} \left| \frac{\sum_{i=1}^n \mathbb{1}(x^i \leq l)}{n+1} - \Phi(s_l) \right| \\ &\leq \frac{1}{\min\left\{\phi\left(\Phi^{-1}\left(\frac{p_1}{2}\right)\right), \phi\left(\Phi^{-1}\left(1 - \frac{p_k}{2}\right)\right)\right\}} \cdot 2(k-1)\epsilon \end{aligned}$$

Let $K_2 = 1/\min\left\{\phi\left(\Phi^{-1}\left(\frac{p_1}{2}\right)\right), \phi\left(\Phi^{-1}\left(1 - \frac{p_k}{2}\right)\right)\right\}$. Adjusting the constants, for $2(k-1)K_2n^{-1} < \epsilon < 2(k-1)K_1K_2$, we have

$$P\left(\|\hat{S} - S\|_1 > \epsilon\right) \leq 2 \exp\left\{-\frac{1}{8K_2^2} \cdot \frac{n\epsilon^2}{(k-1)^2}\right\}.$$

$\square$